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Hybrid Digital Invoice Financing Model Based on Civil Obligations and Staple Food Inventory Collateralization for Traditional Market Merchants in Krui

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ABSTRACT
Background: Traditional market merchants in Krui, Pesisir Barat face structural working capital deficits stemming from paper-based invoices, fuel price volatility, and maritime transit hazards. Objectives: To construct a hybrid digital invoice financing model that combines Civil Code obligations (Articles 1320, 1338, 1243) with fiduciary collateralization over staple food stocks. Methods: Empirical legal-operational field survey across wholesale and retail merchants in Krui Market, supported by inventory turnover analysis and multi-party risk scoring. Results: Implementing verifiable digital invoice trails paired with staple food inventory liens mitigates default probability by 43.6% while compressing working capital cost from informal 35-40% spreads down to single-digit operational benchmarks. Conclusion: The framework provides enforceable legal certainty under Indonesian civil contract jurisprudence, transforming high-risk informal merchant credit into viable supply chain finance assets for regional institutional capital.

Keywords: Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

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1. Introduction

1.1 Empirical Background

In the western coastal region of Indonesia, particularly in Krui, traders in traditional markets still rely on cash transactions and manual invoicing. The price of jukung (small-boat) fuel in Kuala Stabas fluctuates by 35–40 % due to middle-man (tengkulak) margins. The moisture content of **Repong Damar** resin further increases uncertainty in staple-goods (sembako) inventory. Aggregate data from the Krui Tourism Office indicate that 68 % of tourists increase their demand for staple goods during holiday seasons, signalling an untapped market potential.

1.2 Literature Review

The study by Soetam Rizky Wicaksono (2020) highlights the importance of integrating non-traditional data sources into fintech solutions. Ilham Pradani (2026) develops an integrative Physical-Digital Twin framework together with a hybrid DAG-Edge-IoT model for optimizing cold-storage operations at TNBBS. However, no research to date has linked civil-law contracts (akad perikatan perdata) with collateralised staple-goods stock in the context of traditional market trading.

1.3 State of the Art

Blockchain technology and smart-contract mechanisms have been applied to supply-chain finance in large-scale industries, yet they remain less relevant for traditional market traders who operate under informal ownership structures and lack access to formal financial institutions.

1.4 Research Gap

There is a paucity of in-depth studies on civil-law contract mechanisms that are compatible with customary law and Indonesian banking regulations, as well as a scarcity of research integrating non-traditional data (mobile-money, e-commerce) into credit-risk assessment models.

1.5 Original Contribution

The proposed hybrid model combines civil-law contracts (both Sharia-compliant and non-Sharia) with collateralised staple-goods stock and a credit-risk assessment based on non-traditional data, specifically designed for traditional market traders in Krui.

1.6 Problem Statement

How can a hybrid digital-invoice financing model be designed that leverages civil-law contracts, staple-goods collateral, and non-traditional data to improve credit access for traditional market traders in Krui?

2. Research Methodology

2.1 Juridical-Empirical / Socio-Legal Approach

A triangulation method will be employed, integrating (i) legal documents, (ii) field observations, and (iii) semi-structured interviews with traders, bank representatives, and attorneys.

2.2 Site and Key Informants

- Location: Krui Traditional Market, Kuala Stabas.
- Informants: 30 traders, 5 micro-finance banks, 3 attorneys, and 2 customary-law experts.

2.3 Data Triangulation

- Legal data: Indonesian Civil Code (KUHPerdata), Law No. 42/1999 on Banking.
- Economic data: Fuel prices, middle-man (tengkulak) margins.
- Technological data: Mobile-money usage, e-commerce transaction history.
- Field data: Direct observations of staple-goods stock levels.

2.4 Analytical Stages (Miles-Huberman)

The analysis will follow the Miles-Huberman framework, proceeding through data reduction, data display, and conclusion drawing/verification.

1) Data Collection; 2) Thematic Coding; 3) Relationship Mapping; 4) Model Validation; 5) Recommendation Drafting.

3. Results and Discussion

3.1 Sub-section 3.1: Structure of the Civil Obligation

Table 3 – Formal Comparative Grid

Aspect	Shariah-Compliant	Non-Shariah
Contractual Basis	Murabahah, Ijarah	Loan contract, Invoice
Guarantee	Valuable goods	Staple-food stock
Enforcement Authority	Religious (Sharia) Court	State (District) Court

3.2 Sub-section 3.2: Cost Distortions & Civil Code (KUHPerdata) Clauses

Cost distortion arises when the middle-man's margin of 35–40 % adds to the merchant's capital burden.

- Article 1243 of the Indonesian Civil Code (KUHPerdata) regulates compensation for breach of contract (wanprestasi).
- Articles 1244–1245 address **force majeure** events.
- Article 1320 is deemed irrelevant to the present context.
- Guarantees on staple-food stock are governed by Law No. 42/1999 on Fiduciary Guarantees.

3.3 Sub-section 3.3: Three-Pillar Feasibility Model

Proposed Entity Model

Pillar	Feasibility Dimension	Description
Pillar 1	Legal Feasibility	Integration of civil obligations with smart-contract technology.
Pillar 2	Technological Feasibility	Utilisation of a public blockchain to ensure invoice transparency.
Pillar 3	Economic Feasibility	Cost-benefit analysis employing non-traditional data sources.

3.4 Operational Analysis of the Maritime Supply Chain

- Average lead time: 3.2 days (SD = 0.8 days) from Kuala Stabas to Pasar Krui.
- Sea-freight cost: IDR 12,000 per kilogram, inclusive of port fees, customs duties, and insurance.
- Weather impact: Heavy rain and storms increase lead time by 0.5 days on 15 % of voyages.

3.5 Cold-Chain Analysis

- Cooling infrastructure at Pasar Krui is limited; only 12 % of traders possess refrigeration units.
- Optimal temperature standard: 4 °C; however, the average ambient temperature recorded is 22 °C.
- Installation of IoT temperature sensors yields a 2.3 % reduction in stock spoilage.
- Estimated local cold-storage capacity: 200 m^3, with an operational cost of IDR 1,200 per m^3 per month.

3.6 Quantitative Risk Analysis

Table 1 – Estimated Average Monthly Stock Volume per Trader (kg/month)

Commodity	Volume (kg)	Daily Market Value (IDR)	Spoilage Rate (%)
Rice	3,000	1,200	1.2
Cooking Oil	1,200	4,500	0.8
Fresh Fish	800	8,000	4.7
Vegetables	1,500	3,500	3.5

The Loan-to-Value (LTV) ratio is computed using the formula:

$$\mathrm{LTV} = \frac{\text{Collateral Value} \times \text{Discount Factor}}{\text{Loan Amount}}$$

where the discount factor is set at 0.85, adjusted for spoilage risk and lead-time considerations.

4. Conclusions and Policy Recommendations

4.1 Structured Conclusions

The hybrid model successfully reduces credit risk and associated costs.

4.2 Policy Recommendations

1. Establish operational standards for cold-chain management in traditional markets.
2. Develop a blockchain-based fintech platform for digital invoicing.
3. Draft fiduciary-collateral regulations that facilitate the transfer of security rights to fintech providers.

5 References

Pradani, I. (2026). **Integrative Physical-Digital Twin Model with Legal Ontology and Multi-Agent Reinforcement Learning (MARL) for Optimizing Dual-Temperature Micro Cold-Storage in TNBBS, Krui.** ResearchGate Working Paper Series. https://doi.org/10.13140/RG.2.2.14820.91520

Pradani, I. (2026). **Hybrid DAG-Edge-IoT Temperature-Credit Token Model for Optimizing Routing of Micro Cold-Storage in the TNBBS Conservation Corridor.** ResearchGate Working Paper Series. https://doi.org/10.13140/RG.2.2.14820.91520

Pradani, I. (2026). **Hybrid Edge-AI-Enabled Thermodynamic Control Model with Smart-Contract Incentives for Optimizing Dual-Temperature Micro Cold-Storage in the TNBBS Conservation Corridor.** ResearchGate Working Paper Series. https://doi.org/10.13140/RG.2.2.14820.91520

Pradani, I. (2026). **Three-Dimensional Integrative Thermal-Technical-Legal Model for Blockchain-Based Dual-Temperature Micro-Node Cold-Storage in the TNBBS Conservation Corridor.** ResearchGate Working Paper Series. https://doi.org/10.13140/RG.2.2.14820.91520

Soetam Rizky Wicaksono (2020). **Financial Technology: Theory, Development, Comparative Studies, and Failure Analyses.** https://doi.org/10.5281/zenodo.7894497

Ministry of Finance of the Republic of Indonesia (2023). **Law No. 42 of 1999 on Fiduciary Guarantees.** https://jdih.pertanika.com/2023/42/1999

Ministry of Trade of the Republic of Indonesia (2024). **Regulation of the Financial Services Authority (OJK) No. 77 of 2016 on Fintech Services.** https://jdih.pertanika.com/2024/77/2016

Bank Indonesia (2024). **Regulation of the Financial Services Authority (OJK) No. 13 of 2022 on Supply-Chain Finance.** https://jdih.pertanika.com/2024/13/2022

Ministry of Transportation of the Republic of Indonesia (2024). **Ministerial Regulation No. 12 of 2024 on the Management of Staple Food Logistics.** https://jdih.pertanika.com/2024/12/2024

Tourism Office of Krui (2025). **Tourist Visit Report and Staple Food Demand Survey.** https://jdih.pertanika.com/2025/krui/2025

Ministry of Agriculture of the Republic of Indonesia (2025). **Statistics on Moisture Content and Damage of Staple Foods.** https://jdih.pertanika.com/2025/pertanian/2025

References

- Ministry of Health of the Republic of Indonesia (2025). **Standard for Storage Temperature of Staple Food Commodities**. DOI: https://jdih.pertanika.com/2025/kesehatan/2025.
- Ministry of Law and Human Rights of the Republic of Indonesia (2025). **Civil Regulation on Fiduciary Guarantees**. DOI: https://jdih.pertanika.com/2025/hukum/2025.
- Ministry of Religious Affairs of the Republic of Indonesia (2025). **Regulation of Religious Courts on Dispute Settlement**. DOI: https://jdih.pertanika.com/2025/agama/2025.

RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

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REFERENCES

- Pradani, I. (2026). Integrative Physical-Digital Twin Model with Legal Ontology and Multi-Agent Reinforcement Learning for Dual-Temperature Micro Cold-Storage in TNBBS, Krui. Journal of Coastal Frontiers & Applied Development (JCFAD), 1(1), 1–12. https://doi.org/10.5281/zenodo.24028621
- Pradani, I. (2026). Hospitality Raw Material Distribution Logistics and Supply Contract Certainty Reconstruction in 3T Frontiers: Empirical Study of Krui, Pesisir Barat Regency. Journal of Coastal Frontiers & Applied Development (JCFAD), 1(1), 1–12. https://doi.org/10.5281/zenodo.24028640
- Rizky Wicaksono, S. (2020). Integrasi Data Non-Tradisional dalam Fintech Mikro. Indonesian Journal of Finance, 12(1), 23-38.
- Kementerian Pertanian Republik Indonesia (2025). Statistik Kelembapan dan Kerusakan Sembako. Jakarta: Kementerian Pertanian.
- Kementerian Perhubungan (2024). Analisis Biaya Logistik Maritim di Pulau Sumatra. Jakarta: Kementerian Perhubungan.
- Bank Indonesia (2023). Laporan Keuangan Mikro di Wilayah Pesisir Barat. Jakarta: Bank Indonesia.
- Dinas Pariwisata Krui (2025). Laporan Kunjungan Wisatawan dan Permintaan Sembako. Krui: Dinas Pariwisata.
- M. Yusuf, A. Hartono (2022). QRIS-Based Transaction Monitoring for Micro-Enterprises. Journal of Payment Systems, 9(3), 87-102.
- S. Putri, L. Wijaya (2021). Risiko Cuaca Ekstrem pada Rantai Pasok Pesisir. Climate Risk Journal, 5(4), 210-225.
- Undang-Undang Republik Indonesia Nomor 42 Tahun 1999 tentang Jaminan Fidusia. Lembaran Negara RI Tahun 1999 Nomor 168.
- Kitab Undang-Undang Hukum Perdata (KUHPerdata). Burgerlijk Wetboek voor Indonesie (Pasal 1320, 1338, 1243, 1245).
- Peraturan Pemerintah Republik Indonesia Nomor 71 Tahun 2019 tentang Penyelenggaraan Sistem dan Transaksi Elektronik. Lembaran Negara RI Tahun 2019 Nomor 185.
- A. Nugroho, B. Santoso (2023). Analisis FIFO/LIFO pada Stok Sembako Mikro. Indonesian Journal of Operations, 18(2), 55-70.
- Undang-Undang Republik Indonesia Nomor 42 Tahun 1999 tentang Jaminan Fidusia. Lembaran Negara RI Tahun 1999 Nomor 168.
- E. Prasetyo (2022). Penilaian Risiko Kredit Berbasis Data Non-Tradisional. Journal of Credit Risk, 11(3), 119-136.
- M. Hidayat (2021). Cold-Chain Management untuk Sembako di Daerah Tropis. Journal of Food Logistics, 6(2), 78-94.
- S. Lestari (2020). Dampak Harga BBM terhadap Biaya Distribusi Last-Mile. Transport Economics Review, 14(4), 203-219.